```
use sagar_ass9

db.createCollection("student")

db.student.insertMany([
  { name: "Ram", rollNo: 101, marks: 85, CGPA: 9.2, branch: "Computer Science", nativeCity: "Delhi" },
  { name: "Sita", rollNo: 102, marks: 88, CGPA: 8.6, branch: "IT", nativeCity: "Mumbai" },
  { name: "Shyam", rollNo: 103, marks: 75, CGPA: 7.9, branch: "Electrical", nativeCity: "Kolkata" },
  { name: "Ravi", rollNo: 104, marks: 90, CGPA: 9.5, branch: "Computer Science", nativeCity:
"Bangalore" },
  { name: "Priya", rollNo: 105, marks: 80, CGPA: 8.2, branch: "Electronics", nativeCity: "Chennai" },
  { name: "Amit", rollNo: 106, marks: 95, CGPA: 9.8, branch: "Computer Science", nativeCity: "Delhi" },
  { name: "Geeta", rollNo: 107, marks: 78, CGPA: 8.0, branch: "Civil", nativeCity: "Hyderabad" }
])

db.student.find().pretty()

db.student.updateOne(
  { name: "Ram" },
  { $set: { marks: 90 } }
)

db.student.find({
  $or: [{ CGPA: { $gte: 9 } }, { branch: "Computer Science" }]
}).pretty()

db.student.find({ branch: { $ne: "IT" } }).pretty()

db.student.find({ CGPA: 7, nativeCity: "Mumbai" }).pretty()

db.student.find({
  $nor: [{ CGPA: { $eq: 9 } }, { branch: "Computer Science" }]
}).pretty()

db.student.save({
  name: "Nisha",
  rollNo: 108,
  marks: 88,
  CGPA: 8.1,
  branch: "Computer Science",
  nativeCity: "Mumbai"
})
db.student.find().pretty()

db.student.deleteOne({ name: "Sita" })

db.student.find({ branch: "Computer Science" }).sort({ CGPA: -1 }).limit(1).pretty()
```

// ----------

```
// 📘 ADDITIONAL NOTES & COMMENTS (FOR VIVA / REPORT)
// ------------------------------------------------------------
//
// 🔹 DATABASE NAME: sagar_ass9
// 🔹 COLLECTION NAME: student
// 🔹 NUMBER OF DOCUMENTS: 7 (initially), then +1 added (Nisha), -1 deleted (Sita)
//
// ------------------------------------------------------------
// 🔹 PURPOSE:
// This practical demonstrates how to perform all major CRUD operations
// (Create, Read, Update, Delete) in MongoDB along with the use of logical operators.
// ------------------------------------------------------------
//
// 🔹 CRUD OPERATIONS USED:
// ▪ CREATE → insertMany(), save()
// ▪ READ → find(), find().pretty()
// ▪ UPDATE → updateOne()
// ▪ DELETE → deleteOne()
//
// ------------------------------------------------------------
// 🔹 QUERY OPERATORS USED:
// ▪ $or  → Combine multiple conditions (logical OR)
// ▪ $ne  → Not equal to
// ▪ $gte → Greater than or equal to
// ▪ $nor → Negation of multiple conditions
//
// ------------------------------------------------------------
// 🔹 AGGREGATE FEATURES USED:
// ▪ sort({ field: -1 }) → Sort descending (highest CGPA first)
// ▪ limit(1) → Restrict output to one document
//
// ------------------------------------------------------------
// 🔹 SAMPLE OUTPUT EXPECTED:
//
// ➤ db.student.find().pretty()
// { name: "Ram", rollNo: 101, marks: 90, CGPA: 9.2, branch: "Computer Science", nativeCity: "Delhi" }
// { name: "Amit", rollNo: 106, marks: 95, CGPA: 9.8, branch: "Computer Science", nativeCity: "Delhi" }
// ...
//
// ➤ db.student.find({ branch: "Computer Science" }).sort({ CGPA: -1 }).limit(1)
// { name: "Amit", rollNo: 106, marks: 95, CGPA: 9.8, branch: "Computer Science", nativeCity: "Delhi" }
//
// ------------------------------------------------------------
// 🔹 VIVA SHORT ANSWERS:
//
// Q1️⃣: What is MongoDB?
// ➤ MongoDB is a NoSQL, document-oriented database that stores data in JSON-like BSON format.
//
// Q2️⃣: What is a collection?
// ➤ A collection in MongoDB is similar to a table in SQL. It stores multiple documents.
//
// Q3️⃣: What are CRUD operations?
// ➤ CRUD stands for Create, Read, Update, and Delete — the four basic database operations.
//
// Q4️⃣: Difference between find() and pretty()?
```

// ➤ find() shows raw JSON; pretty() displays formatted and indented results.
//
// Q 5 : How is data sorted in MongoDB?
// ➤ Using sort({ field: 1 }) for ascending and sort({ field: -1 }) for descending order.
//
// Q 6 : Why use $or and $nor operators?
// ➤ $or finds documents matching at least one condition,
// while $nor excludes documents matching any of the listed conditions.
//
// Q 7 : What does save() do in MongoDB?
// ➤ save() either inserts a new document or updates an existing one if _id already exists.
//
// ----------
// 🔹 REAL-LIFE APPLICATION EXAMPLE:
// ▪ Used in student record management systems to store marks, CGPA, and course details.
// ▪ MongoDB allows flexible schema, making it ideal for college data, e-commerce, or IoT apps.
//
// ----------
// 🔹 CONCLUSION:
// This assignment demonstrates efficient data handling in MongoDB using shell commands,
// showing how NoSQL databases provide flexible and schema-free document storage with
// simple yet powerful query syntax.